

# ANTHONY LIAO

Software Engineer

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## SKILLS

**Languages:** C# · Python · R · SQL · JavaScript · Java · C++ · Mandarin

**Tech Stack:** .NET · ASP.NET Core · Blazor · WPF · Azure DevOps · SQL Server · Dapper · Git · xUnit

**Tools & Data:** Visual Studio · Pandas · NumPy · Matplotlib · CatBoost · SHAP · Power BI · Tableau · OpenCV

## EXPERIENCE

**Software Engineer · NIAR – Advanced Materials Research**

**Aug 2024 – Present | Wichita, KS**

- Co-led development of PATH, an enterprise web application that replaced a legacy Access database with a modern system for material tracking, machine scheduling, and quality control.
- Designed the backend architecture and wrote recursive SQL queries to trace the full history of how raw materials were blended, reused, and distributed across manufacturing builds — giving engineers complete material genealogy at a glance.
- Wrapped complex multi-step database operations in transactions so that inventory updates, build logs, and material recovery either completed fully or rolled back cleanly, preventing any partial or corrupted saves.
- Built a defect reporting system where flagging a build automatically flags all its child parts, and a maintenance scheduler that tracks when each machine is due for service based on time or usage.
- Built frontend workflows including a visual material lineage graph and an Excel-style data grid using Handsontable and JS Interop for high-volume part entry directly in the browser.
- Used QuestPDF to automatically generate formatted multi-page build summary reports, giving engineers a clean downloadable record at the end of each production run.
- Built MARS, a desktop tool suite where new analysis modules can be plugged in independently — automating the Excel-heavy data reduction work that engineers previously did by hand after every test.
- Created internal scripts and tools that automated repetitive data tasks across the lab, saving engineers significant manual work each week.

**Software Engineer · NIAR – ATLAS**

**Aug 2021 – Aug 2024 | Wichita, KS**

- Worked on a C# / .NET manufacturing system covering data acquisition, defect tracking, and WPF interface improvements — integrated with an OpenCV image analysis engine and SQL Server for inspection reporting.

## RESEARCH

**Data Science Collaborator · Independent**

**2024 – Present | Wichita, KS**

- Partnered with medical researchers on statistical analysis, data visualization, and predictive modeling — one paper published and one under review.

## PROJECTS

**Predicting Unmet Healthcare Needs Among Adults with Disabilities / R · CatBoost · SHAP · targets**

- Built a survey-weighted machine learning pipeline on CDC BRFSS data (3.9M adults, 2016-2024) to predict unmet healthcare needs among adults with disabilities — comparing CatBoost, Random Forest, and logistic regression with SHAP explainability and subgroup fairness audits across race, income, and disability type.

**QUEST: Quiz Understanding & Evaluation for Study Tracking / C# · WPF**

- Independently built a desktop app that analyzes exported quiz results, identifies weak subject areas, and gives actionable study recommendations.

## EDUCATION

**M.S. in Data Science**

**Wichita State University · May 2026**

**Relevant Coursework:** Machine Learning, Data Visualization, Artificial Intelligence, Advanced Data Storage.

**B.S. in Computer Science · Magna Cum Laude · 3.7 GPA**

**Wichita State University · May 2024**

**Relevant Coursework:** Data Structures, Algorithms, Database Systems, Software Engineering, Operating Systems.